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MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

Argument

$$\log_b a$$

Base

Study Material

Logarithm

(English Medium)

LOGARITHM

1. DEFINITION :

The logarithm of a number N to a base 'a' is an exponent indicating the power to which the base 'a' must be raised to obtain the number N. This number is designated as $\log_a N$. (Read it "Log N on base a"). Here N is usually called argument of the Logarithm and 'a' is called base of the Logarithm.

Hence : $\log_a N = x \Leftrightarrow a^x = N$, $a > 0$, $a \neq 1$ and $N > 0$

By the definition of logarithm, $\log_2 16$ is the exponent indicating the power to which 2 must be raised in order to obtain 16.

As $2^4 = 16$, hence $\log_2 16 = 4$.

Similarly

Exponential Form		Logarithmic Form
$3^5 = 243$	$\Leftrightarrow$	$\log_3 243 = 5$
$5^4 = 625$	$\Leftrightarrow$	$\log_5 625 = 4$
$2^{-3} = \frac{1}{8}$	$\Leftrightarrow$	$\log_2 \frac{1}{8} = -3$
$7^0 = 1$	$\Leftrightarrow$	$\log_7 1 = 0$

Note that the expressions $\log_3(-27)$, $\log_1 16$, $\log_0 5$ and $\log_2 0$ has no sense in real numbers since the equations $3^x = -27$, $1^x = 16$, $0^x = 5$, $2^x = 0$ are absurd for any real x, the reason being obvious that no such exponent x in real number could be found.

In general, the expression $\log_a N$ is meaningful if and only if, $a > 0$, $a \neq 1$ and $N > 0$.

The existence and uniqueness of the number $\log_a N$ follows from the properties of exponential functions.

Illustration 1 : If $\log_4 m = 1.5$, then find the value of m.

Solution : $\log_4 m = 1.5 \Rightarrow m = 4^{3/2} \Rightarrow m = 8$

Illustration 2 : If $\log_5 p = a$ and $\log_2 q = a$, then prove that $\frac{p^4 q^4}{100} = 100^{2a-1}$

Solution : $\log_5 p = a \Rightarrow p = 5^a$

$\log_2 q = a \Rightarrow q = 2^a$

$$\Rightarrow \frac{p^4 q^4}{100} = \frac{5^{4a}.2^{4a}}{100} = \frac{(10)^{4a}}{100} = \frac{(100)^{2a}}{100} = 100^{2a-1}$$

Illustration 3 : The value of N, satisfying $\log_a[1 + \log_b\{1 + \log_c(1 + \log_p N)\}] = 0$ is -

(A) 4 (B) 3 (C) 2 (D) 1

Solution : $1 + \log_b\{1 + \log_c(1 + \log_p N)\} = a^0 = 1$

$\Rightarrow \log_b\{1 + \log_c(1 + \log_p N)\} = 0 \quad \Rightarrow \quad 1 + \log_c(1 + \log_p N) = 1$

$\Rightarrow \log_c(1 + \log_p N) = 0 \quad \Rightarrow \quad 1 + \log_p N = 1$

$\Rightarrow \log_p N = 0 \quad \Rightarrow \quad N = 1$ **Ans. (D)**

Do yourself - 1 :

1 Express the following in logarithmic form :

(a) $81 = 3^4$ (b) $0.001 = 10^{-3}$ (c) $2 = 128^{1/7}$

2. Express the following in exponential form :

(a) $\log_2 32 = 5$ (b) $\log_{\sqrt{2}} 4 = 4$ (c) $\log_{10} 0.01 = -2$

3. If $\log_{2\sqrt{3}} 1728 = x$, then find x.

4. Find the logarithms of the following numbers to the base 2 :

(i) 1 (ii) 2 (iii) 4 (iv) 8

(v) $\frac{1}{2}$ (vi) $\frac{1}{32}$ (vii) $\frac{1}{16}$ (viii) $\sqrt{2}$

(ix) $\sqrt[3]{8}$ (x) $2\sqrt{2}$ (xi) $\frac{1}{\sqrt[5]{2}}$ (xii) $\frac{1}{\sqrt[7]{8}}$

5. Find the logarithms of the following numbers to the base $\frac{1}{2}$:

(i) 1 (ii) $\frac{1}{2}$ (iii) $\frac{1}{8}$ (iv) 16

(v) $\sqrt{2}$ (vi) $\frac{1}{\sqrt{2}}$ (vii) $2\sqrt{2}$ (viii) $\frac{1}{4\sqrt[4]{2}}$

6. Find the logarithms of the following numbers to the base 3 :

(i) 1 (ii) 3 (iii) 9 (iv) 81

(v) $\frac{1}{3}$ (vi) $\sqrt{3}$ (vii) $\frac{1}{3\sqrt{3}}$ (viii) $27\sqrt{3}$

(ix) $\sqrt[7]{9}$

7. Find the logarithms of the following numbers to the base $\frac{1}{3}$:

(i) 1 (ii) $\frac{1}{3}$ (iii) $\frac{1}{9}$ (iv) 3

(v) 9 (vi) 81 (vii) $\sqrt[3]{3}$ (viii) $\frac{1}{\sqrt[7]{3}}$

(ix) $9\sqrt{3}$ (x) $\frac{1}{9\sqrt[4]{3}}$

8. Find the logarithms of the following numbers to the base 5.

(i) 1 (ii) 5 (iii) 25 (iv) 625

(v) $\frac{1}{5}$ (vi) $\frac{1}{25}$ (vii) $\frac{1}{\sqrt{5}}$ (viii) $\sqrt{\sqrt{5}}$

(ix) $5^{\frac{1}{2}}$ (x) $5^{\frac{1}{3}}$ (xi) $\sqrt[4]{5\sqrt[3]{5}}$

9. Find all values of 'a' for which each of the following equalities hold true :

(i) $\log_2 a = 2$ (ii) $\log_a 2 = 1$ (iii) $\log_a 1 = 0$ (iv) $\log_{10}(a(a+3)) = 1$

(v) $\log_{1/3}(a^2-1) = -1$ (vi) $\log_2(a^2-5) = 2$ (vii) $\log_3 a = 2$ (viii) $\log_{1/3}(a) = 4$

(ix) $\log_{1/3}(a) = 0$ (x) $\log_a(a+2) = 2$ (xi) $\log_3(a^2+1) = 1$

10. Evaluate the following :

(i) $\log_5\left(\frac{1}{\sqrt{5}}\right)$ (ii) $\log_{\sqrt{5}+1}(6+2\sqrt{5})$

(iii) $\log_3\left(4\sin^2(x)+4\cos^2(x)-1\right)$ (iv) $\log_{\sqrt{3}-\sqrt{2}}\left(\sqrt{5-2\sqrt{6}}\right)$

(v) $\log_{1/3}\sqrt[4]{729.\sqrt[3]{9^{-1}}.27^{-4/3}}$

2. FUNDAMENTAL LOGARITHMIC IDENTITY :

From the definition of the logarithm of the number N to the base 'a' , we have an identity :

$$a^{\log_a N} = N \ , \ a > 0 \ , \ a \neq 1 \ \text{ and } \ N > 0$$

This is known as the **Fundamental Logarithmic Identity .**

Note :

Using the basic definition of logarithm we have 3 important deductions :

(a) $\log_a 1 = 0$ i.e. logarithm of unity to any base is zero ($a > 0$; $a \neq 1$).

(b) $\log_N N = 1$ i.e. logarithm of a number to the same base is 1. ($N > 0$; $N \neq 1$)

(c) $\log_{\frac{1}{N}} N = -1 = \log_N \frac{1}{N}$ i.e. logarithm of a number to the base as its reciprocal is –1. ($N > 0$; $N \neq 1$)

3. THE PRINCIPAL PROPERTIES OF LOGARITHMS :

If m,n are arbitrary positive numbers where $a > 0$, $a \neq 1$ and x is any real number, then

(a) $\log_a mn = \log_a m + \log_a n$ (b) $\log_a \frac{m}{n} = \log_a m - \log_a n$ (c) $\log_a m^x = x \log_a m$

Illustration 4 : Find the value of $2\log\frac{2}{5} + 3\log\frac{25}{8} - \log\frac{625}{128}$

Solution :

$$2\log\frac{2}{5} + 3\log\frac{25}{8} + \log\frac{128}{625}$$

$$= \log\frac{2^2}{5^2} + \log\left(\frac{5^2}{2^3}\right)^3 + \log\frac{2^7}{5^4}$$

$$= \log\frac{2^2}{5^2}\cdot\frac{5^6}{2^9}\cdot\frac{2^7}{5^4} = \log 1 = 0$$

Illustration 5 : If $\log_e x - \log_e y = a$, $\log_e y - \log_e z = b$ & $\log_e z - \log_e x = c$, then find the value of

$$\left(\frac{x}{y}\right)^{b-c} \times \left(\frac{y}{z}\right)^{c-a} \times \left(\frac{z}{x}\right)^{a-b}.$$

Solution :

$$\log_e x - \log_e y = a \Rightarrow \log_e \frac{x}{y} = a \Rightarrow \frac{x}{y} = e^a$$

$$\log_e y - \log_e z = b \Rightarrow \log_e \frac{y}{z} = b \Rightarrow \frac{y}{z} = e^b$$

$$\log_e z - \log_e x = c \Rightarrow \log_e \frac{z}{x} = c \Rightarrow \frac{z}{x} = e^c$$

$$\therefore \left(e^a\right)^{b-c} \times \left(e^b\right)^{c-a} \times \left(e^c\right)^{a-b}$$

$$= e^{a(b-c)+b(c-a)+c(a-b)} = e^0 = 1$$

Illustration 6 : If $a^2 + b^2 = 23ab$, then prove that $\log\frac{(a+b)}{5} = \frac{1}{2}(\log a + \log b)$.

Solution : $a^2 + b^2 = (a + b)^2 - 2ab = 23ab$

$\Rightarrow (a + b)^2 = 25ab \Rightarrow a+b = 5\sqrt{ab}$ ....(i)

Using (i)

L.H.S. $= \log\frac{(a+b)}{5} = \log\frac{5\sqrt{ab}}{5} = \frac{1}{2}\log ab = \frac{1}{2}(\log a + \log b) =$ R.H.S.

Illustration 7 : If $\log_a x = p$ and $\log_b x^2 = q$, then $\log_x \sqrt{ab}$ is equal to (where $a, b, x \in R^+ - \{1\}$)-

(A) $\frac{1}{p}+\frac{1}{q}$ (B) $\frac{1}{2p}+\frac{1}{q}$ (C) $\frac{1}{p}+\frac{1}{2q}$ (D) $\frac{1}{2p}+\frac{1}{2q}$

Solution : $\log_a x = p \Rightarrow a^p = x \Rightarrow a = x^{1/p}$.

similarly $b^q = x^2 \Rightarrow b = x^{2/q}$

Now, $\log_x \sqrt{ab} = \log_x \sqrt{x^{1/p}x^{2/q}} = \log_x x^{\left(\frac{1}{p}+\frac{2}{q}\right)\cdot\frac{1}{2}} = \frac{1}{2p}+\frac{1}{q}$

Do yourself - 2 :

1 Find the value of the following :

(a) $\log_{1.4\bar{3}} \frac{43}{30}$ **(b)** $\left(\frac{1}{2}\right)^{\log_2 5}$

2. If $4^{\log_2 2x} = 36$, then find x.

3. Show that $\frac{1}{2}\log 9 + 2\log 6 + \frac{1}{4}\log 81 - \log 12 = 3\log 3$

4. Find the value of following

(i) $\log_{12} 8 + \log_{12} 3 + \log_{12} 6$ (ii) $\log_5 \frac{500}{3} - \log_5 \frac{4}{3}$

(iii) $\log_{39} \frac{15}{7} + \log_{39} \frac{13}{3} - \log_{39} \frac{5}{21}$ (iv) $2\log_6 2 + 3\log_6 3 + \log_6 12$

4. BASE CHANGING THEOREM :

Can be stated as "quotient of the logarithm of two numbers is independent of their common base."

Symbolically, $\log_b m = \frac{\log_a m}{\log_a b}$, where $a > 0,\ a \neq 1,\ b > 0,\ b \neq 1$

Note :

(i) $\log_b a.\ \log_a b = \frac{\log a}{\log b}.\frac{\log b}{\log a} = 1$; hence $\boxed{\log_b a = \frac{1}{\log_a b}}$.

(ii) $a^{\log_b c} = c^{\log_b a}$

(iii) **Base power formula :** $\log_{a^k} m = \frac{1}{k}\log_a m$

(iv) The base of the logarithm can be any positive number other than 1, but in normal practice, only two bases are popular, these are 10 and e(=2.718 approx). Logarithms of numbers to the base 10 are named as 'common logarithm' and the logarithms of numbers to the base e are named as Natural or Napierian logarithm. **We will consider log x as $\log_e$ x or ℓn x.**

(v) Conversion of base e to base 10 & viceversa :

$$\log_e a = \frac{\log_{10} a}{\log_{10} e} = 2.303 \times \log_{10} a \; ; \quad \log_{10} a = \frac{\log_e a}{\log_e 10} = \log_{10} e \times \log_e a = 0.434 \log_e a$$

(vi) Some important values : $\log_{10}2 \approx 0.3010$; $\log_{10}3 \approx 0.4771$; $\ell n2 \approx 0.693$, $\ell n10 \approx 2.303$

(vii) The positive real number 'n' is called the antilogarithm of a number 'm' to base 'a' if $\log_a n = m$

Thus, $\log_a n = m \Leftrightarrow n = \text{antilog}_a\, m$

Illustration 8 : If a, b, c are distinct positive real numbers different from 1 such that $(\log_b a \,.\, \log_c a - \log_a a) + (\log_a b \,.\, \log_c b - \log_b b) + (\log_a c \,.\, \log_b c - \log_c c) = 0$, then abc is equal to -

(A) 0 (B) e (C) 1 (D) none of these

Solution : $(\log_b a \log_c a - 1) + (\log_a b \,.\, \log_c b - 1) + (\log_a c \log_b c - 1) = 0$

$$\Rightarrow \frac{\log a}{\log b}.\frac{\log a}{\log c} + \frac{\log b}{\log a}.\frac{\log b}{\log c} + \frac{\log c}{\log a}.\frac{\log c}{\log b} = 3$$

$\Rightarrow (\log a)^3 + (\log b)^3 + (\log c)^3 = 3\log a \log b \log c$

$\Rightarrow (\log a + \log b + \log c) = 0$ [$\because$ If $a^3 + b^3 + c^3 - 3abc = 0$, then $a + b + c = 0$ if $a \neq b \neq c$]

$\Rightarrow \log abc = \log 1 \Rightarrow abc = 1$

Illustration 9 : Evaluate : $81^{1/\log_5 3} + 27^{\log_9 36} + 3^{4/\log_7 9}$

Solution : $81^{\log_3 5} + 3^{3\log_9 36} + 3^{4\log_9 7}$

$= 3^{4\log_3 5} + 3^{\log_3 (36)^{3/2}} + 3^{\log_3 7^2} = 625 + 216 + 49 = 890.$

Illustration 10 : Show that $\log_4 18$ is an irrational number.

Solution : $\log_4 18 = \log_4(3^2 \times 2) = 2\log_4 3 + \log_4 2 = 2\dfrac{\log_2 3}{\log_2 4} + \dfrac{1}{\log_2 4} = \log_2 3 + \dfrac{1}{2}$

assume the contrary, that this number $\log_2 3$ is rational number.

$\Rightarrow \quad \log_2 3 = \dfrac{p}{q}$. Since $\log_2 3 > 0$ both numbers p and q may be regarded as natural number

$\Rightarrow \quad 3 = 2^{p/q} \quad \Rightarrow \quad 2^p = 3^q$

But this is not possible for any natural number p and q. The resulting contradiction completes the proof.

Illustration 11 : If in a right angled triangle, a and b are the lengths of sides and c is the length of hypotenuse and $c-b \neq 1$, $c+b \neq 1$, then show that

$\log_{c+b}a + \log_{c-b}a = 2\log_{c+b}a \,.\, \log_{c-b}a.$

Solution : We know that in a right angled triangle

$c^2 = a^2 + b^2$

$c^2 - b^2 = a^2$ .... (i)

$$\text{LHS} = \frac{1}{\log_a(c+b)} + \frac{1}{\log_a(c-b)} = \frac{\log_a(c-b)+\log_a(c+b)}{\log_a(c+b).\log_a(c-b)}$$

$$= \frac{\log_a(c^2-b^2)}{\log_a(c+b).\log_a(c-b)} = \frac{\log_a a^2}{\log_a(c+b).\log_a(c-b)} \qquad \text{(using (i))}$$

$$= \frac{2}{\log_a(c+b).\log_a(c-b)} = 2\log_{(c+b)}a \,.\, \log_{(c-b)}a = \text{RHS}$$

Do yourself - 3 :

1. **(i)** Evaluate : $\dfrac{\log_3 135}{\log_{15} 3} - \dfrac{\log_3 5}{\log_{405} 3}$ **(ii)** Evaluate : $\log_9 27 - \log_{27} 9$

(iii) Evaluate : $2^{\log_3 5} - 5^{\log_3 2}$ **(iv)** Evaluate : $\log_3 4 . \log_4 5 . \log_5 6 . \log_6 7 . \log_7 8 . \log_8 9$

(v) If $\dfrac{1}{\log_3 \pi} + \dfrac{1}{\log_4 \pi} > x$, then x can be -

(A) 2 (B) 3 (C) 3.5 (D) π

(vi) If $\log_a 3 = 2$ and $\log_b 8 = 3$, then $\log_a b$ is -

(A) $\log_3 2$ (B) $\log_2 3$ (C) $\log_3 4$ (D) $\log_4 3$

2. Find value of following :

(i) $\log_7 3 . \log_5 2 . \log_3 7 . \log_2(125)$ (ii) $25^{\log_5 3}$

(iii) $6^{\log_6 5} + 3^{\log_9 16}$ (iv) $\log_6 4 + \dfrac{1}{\log_9 6}$

3. Given that $\log_2 a = s$, $\log_4 b = s^2$ and $\log_{c^2}(8) = \dfrac{2}{s^3+1}$. Write $\log_2 \dfrac{a^2b^5}{c^4}$ as a function of 's' (a, b, c > 0, c ≠ 1).

4. Prove that $a^x - b^y = 0$ where $x = \sqrt{\log_a b}$ & $y = \sqrt{\log_b a}$, $a > 0$, $b > 0$ & $a , b \neq 1$.

5. Compute the following : $a^{\frac{\log_b(\log_b N)}{\log_b a}}$

6. The value of $\log_5 9 . \log_2(\sqrt{5}+2) . \log_{(\sqrt{5}-2)} 5 . \log_3\left(\frac{1}{4}\right)$ is

7. Calculate : $4^{5\log_{4\sqrt{2}}\left(3-\sqrt{6}\right)-6\log_8\left(\sqrt{3}-\sqrt{2}\right)}$

8. Simplify : $\frac{81^{\frac{1}{\log_5 9}}+3^{\frac{3}{\log_{\sqrt{6}} 3}}}{409}.\left(\left(\sqrt{7}\right)^{\frac{2}{\log_{25} 7}}-\left(125\right)^{\log_{25} 6}\right)$

9. Simplify : $5^{\log_{1/5}\left(\frac{1}{2}\right)}+\log_{\sqrt{2}}\frac{4}{\sqrt{7}+\sqrt{3}}+\log_{1/2}\frac{1}{10+2\sqrt{21}}$

10. Find the value of $49^{(1-\log_7 2)}+5^{-\log_5 4}$.

11. Find the value of the expression $\frac{2}{\log_4(2000)^6}+\frac{3}{\log_5(2000)^6}$

12. Let $x = \left(\log_{1/3} 5\right)\left(\log_{125} 343\right)\left(\log_{49} 729\right)$ and $y = 25^{3\log_{289} 11\ \log_{28}\sqrt{17}\ \log_{1331} 784}$, then the value of $x^2 + y^2$ is

13. Find antilog of $\frac{5}{6}$ to the base 64.

5. LOGARITHMIC EQUATIONS

Illustration 12 : $\log_{\frac{1}{2}}\left(\log_2\sqrt{2}x\right)=1$, then find x ?

Solution : $\log_{\frac{1}{2}}\left(\log_2\sqrt{2}x\right)=1$

$\Rightarrow \log_2\left(\sqrt{2}x\right)=\frac{1}{2}$

$\Rightarrow \sqrt{2}x = 2^{\frac{1}{2}}$

$\Rightarrow x = 1$

Illustration 13 : Solve the equation $2\log_2(\log_2 x)+\log_{1/2}\left(\frac{3}{2}+\log_2 x\right)=1$.

Solution : Let $\log_2 x = t$

$$\Rightarrow 2\log_2(t)+\log_{\frac{1}{2}}\left(\frac{3}{2}+t\right)=1$$

$$\Rightarrow 2\log_2 t-\log_2\left(\frac{3}{2}+t\right)=1$$

$$\Rightarrow \log_2\left(\frac{t^2}{\frac{3}{2}+t}\right)=1$$

$$\Rightarrow \frac{2t^2}{3+2t}=2$$

$\Rightarrow t^2-2t-3=0$

$\Rightarrow (t+1)(t-3)=0$

$\Rightarrow t=3 \because t>0$

$\Rightarrow x=8$

Illustration 14 : Solve the equation $\log x^2-\log(2x)=3\log 3-\log 6$.

Solution : $\log x^2-\log 2x=3\log 3-\log 6,\ x>0$

$\Rightarrow 2\log x-\log 2-\log x=3\log 3-\log 2-\log 3 \Rightarrow \log x=2\log 3 \Rightarrow \log x=\log 9$

$\Rightarrow x=9$

Illustration 15 : Solve the equation $(\log_5 x)^2+\log_5 x+1=\frac{7}{\log_5 x-1}$

Solution : Put $\log_5 x = t$, we get $t^2+t+1=\frac{7}{t-1}$

$(t-1)(t^2+t+1)=7 \Rightarrow t^3+t^2+t-t^2-t-1=7 \Rightarrow t^3-8=0$

$\Rightarrow (t-2)(t^2+2t+4)=0 \Rightarrow t-2=0;\ t^2+2t+4\neq 0 \Rightarrow t=2$

Now, $t=\log_5 x$, so $\log_5 x=2$

$x=5^2 \Rightarrow x=25$

Illustration 16 : Solve the equation $|x-1|^{\log_2 x^2 - 2\log_x 4} = (x-1)^7$

Solution : Obviously $x = 2$ is a solution. Since, left side is positive, $x - 1 > 0$.

The equation reduces $\log_2 x^2 - 2\log_x 4 = 7$

$$\Rightarrow 2t - \frac{4}{t} = 7, t = \log_2 x$$

$$\Rightarrow 2t^2 - 7t - 4 = 0 \Rightarrow t = 4, -\frac{1}{2}$$

But $t > 0$ since $x > 1$. $\therefore$ $t = 4$

$\Rightarrow x = 2^4 = 16$

$\therefore$ $x = 2,16$

Illustration 17 : Solve the equation $x^{\log_3 x^2 + (\log_3 x)^2 - 10} = \frac{1}{x^2}$

Solution : Taking $\log_3$ on both sides, we get

$(2t + t^2 - 10)t = -2t$, $\quad t = \log_3 x$

$\Rightarrow t(t^2 + 2t - 8) = 0 \Rightarrow t = 0,2,-4$

$$\Rightarrow x = 1, 9, \frac{1}{81}$$

Illustration 18 : Solve the equation $4^{\log_2 \ell nx} = \ell nx - (\ell nx)^2 + 1$

Solution : $4^{\log_2 \ell nx} = 2^{2\log_2(\ell nx)} = 2^{\log_2(\ell nx)^2} = (\ell nx)^2$

$$\Rightarrow (\ell n\, x)^2 = \ell n\, x - (\ell n\, x)^2 + 1 \Rightarrow 2(\ell n\, x)^2 - \ell n\, x - 1 = 0 \Rightarrow \ell n\, x = 1, -\frac{1}{2}$$

But $\ell nx > 0$

$\therefore$ $\ell nx = 1 \Rightarrow x = e$.

Illustration 19 : Solve the equation x : $\log_{x+1}(x^2 + x - 6)^2 = 4$

Solution : We have,

$\log_{x+1}(x^2 + x - 6)^2 = 4 \Rightarrow (x^2 + x - 6)^2 = (x + 1)^4 = (x^2 + 2x + 1)^2$

$\Rightarrow (x^2 + x - 6 - x^2 - 2x - 1)(x^2 + x - 6 + x^2 + 2x + 1) = 0$

$\Rightarrow (-x - 7)(2x^2 + 3x - 5) = 0 \Rightarrow (x + 7)(x - 1)(2x + 5) = 0 \Rightarrow x = -7, -5/2, 1$

The values $x = -7$ and $x = -5/2$ are rejected because they make the base $x + 1$ negative

Hence, $x = 1$ is the only solution of the given equation.

Do yourself - 4 :

1. Find all values of x for which the following equalities hold true.

(i) $\log_2 x^2 = 1$ (ii) $\log_3 x = \log_3(2-x)$ (iii) $\log_4 x^2 = \log_4 x$

(iv) $\log_{1/2}(2x+1) = \log_{1/2}(x+1)$ (v) $\log_{1/3}(x^2+8) = -2$

2. Find all the values of x for which the following equalities hold true.

(i) $\log_2 x^2 = 2$ (ii) $\log_{1/4} x^2 = 1$

(iii) $\log_{1/2} x - \log_{1/2}(3-x) = 0$ (iv) $\log_2(x+1) - \log_2(2x-3) = 0$

Solve the following equations :

3. $\log_3(4^x + 15.2^x + 27) - 2\log_3(4.2^x - 3) = 0$

4. $\dfrac{\log_8\left(8/x^2\right)}{\left(\log_8 x\right)^2} = 3$

5. $\log_x 3.\log_{x/3} 3 + \log_{x/81} 3 = 0$

6. $1 + 2\log_x 2.\ \log_4(10-x) = 2/\log_4 x$

7. $x^{x+1} = x$

8. $x^{\log_x (x+3)^2} = 16$

9. $\sqrt{x^{\log_{10}\sqrt{x}}} = 10$

10. $3^{\log_a x} + 3.x^{\log_a 3} = 2$

11. $\log_a\left(1-\sqrt{1+x}\right) = \log_{a^2}\left(3-\sqrt{1+x}\right)$

12. $2\log_8(2x) + \log_8(x^2+1-2x) = 4/3$

13. $|\log_2 x| = 3$

14. $\log_{x-1} 3 = 2$

15. $\log_4(2\log_3(1+\log_2(1+3\log_3 x))) = \dfrac{1}{2}$

16. $\log_3(1+\log_3(2^x-7)) = 1$

17. $\log_3(3^x - 8) = 2 - x$

18. $\dfrac{\log_2(9-2^x)}{3-x} = 1$

19. $\log_{5-x}(x^2 - 2x + 65) = 2$

20. $\log_{10}5 + \log_{10}(x+10) - 1 = \log_{10}(21x-20) - \log_{10}(2x-1)$

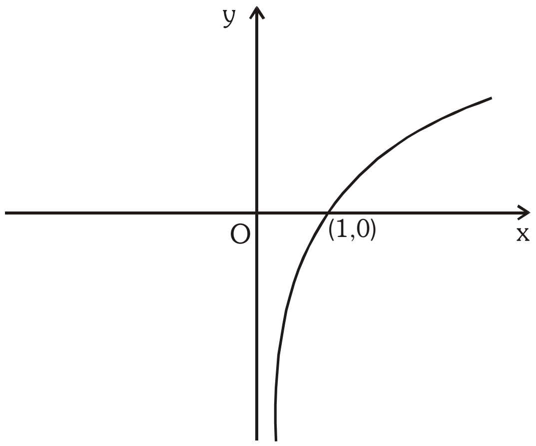
21. $x^{1+\log_{10} x} = 10x$

22. $2(\log_x \sqrt{5})^2 - 3\log_x \sqrt{5} + 1 = 0$

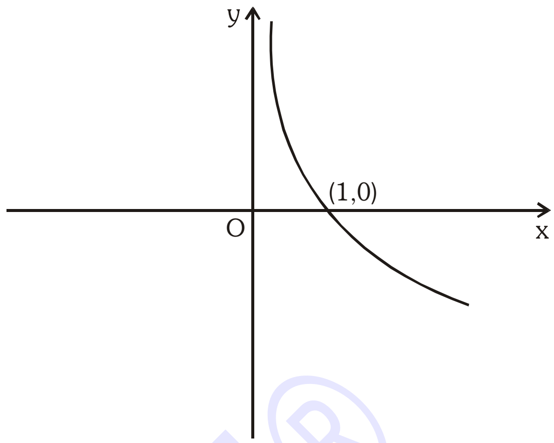
23. $3 + 2\log_{x+1} 3 = 2\log_3(x+1)$

6. GRAPHS OF LOGARITHMIC FUNCTION AND ITS INEQUALITIES

Graph of $y = \log_a x$:



when a > 1



When 0 < a < 1

(i) $\log_a x < \log_a y \Leftrightarrow \begin{cases} x < y & \text{if} \quad a > 1 \\ x > y & \text{if} \quad 0 < a < 1 \end{cases}$

(ii) If a > 1, then $\log_a x < p \Rightarrow 0 < x < a^p$ and $\log_a x > p \Rightarrow x > a^p$

(iii) If 0 < a < 1, then $\log_a x < p \Rightarrow x > a^p$ and $\log_a x > p \Rightarrow 0 < x < a^p$

Note :

(i) If base of logarithm is greater than 1 then logarithm of greater number is greater. i.e. $\log_2 8 = 3$, $\log_2 4 = 2$ etc. and if base of logarithm is between 0 and 1 then on that base logarithm of greater number is smaller. i.e. $\log_{1/2} 8 = -3$, $\log_{1/2} 4 = -2$ etc.

(ii) It must be noted that whenever the number and the base are on the same side of unity then logarithm of that number to that base is positive, however if the number and the base are located on different side of unity then logarithm of that number to that base is negative.

e.g. $\log_{10}\sqrt[3]{10} = \frac{1}{3}$; $\log_{\sqrt{7}} 49 = 4$; $\log_{\frac{1}{2}}\left(\frac{1}{8}\right) = 3$; $\log_2\left(\frac{1}{32}\right) = -5$; $\log_{10}(0.001) = -3$

Illustration 20 : Solve for x : $x^{\log_5 x} > 5$

Solution : as x > 0 (for existance)

now solving inequality

$x^{\log_5 x} > 5$. Taking 'log' with base '5' we have $\log_5 x . \log_5 x > 1$

$\Rightarrow (\log_5 x - 1)(\log_5 x + 1) > 0 \Rightarrow \log_5 x > 1$ or $\log_5 x < -1$

$\Rightarrow x > 5$ or $x < 1/5$. Also we must have $x > 0$

Thus, $x \in (0, 1/5)$ or $x \in (5, \infty)$

Illustration 21 : Solve for x : $\log_3(2x+1) < \log_3 5$.

Solution : Checking existance

$$2x + 1 > 0 \Rightarrow x > -\frac{1}{2}$$

Now solving inequality we have $2x + 1 < 5$

$\Rightarrow \quad 2x < 4$

$\Rightarrow \quad x < 2$

$\Rightarrow \quad x \in (-1/2, 2)$

Illustration 22 : Solve for x : $(\log_{10}100x)^2 + (\log_{10}10x)^2 + \log_{10}x \leq 14$.

Solution : Checking existance

$x > 0$

Now solving inequality,

Let $u = \log_{10}x$

$(2+u)^2 + (1+u)^2 + u \leq 14 \Rightarrow u^2 + 4u + 4 + u^2 + 2u + 1 + u \leq 14$

$\Rightarrow 2u^2 + 7u - 9 \leq 0 \Rightarrow 2u^2 + 9u - 2u - 9 \leq 0$

$\Rightarrow u(2u+9) - 1(2u+9) \leq 0 \Rightarrow (2u+9)(u-1) \leq 0$

$$\Rightarrow \frac{-9}{2} \leq u \leq 1 \Rightarrow \frac{-9}{2} \leq \log_{10} x \leq 1 \Rightarrow 10^{-\frac{9}{2}} \leq x \leq 10$$

Illustration 23 : Solve for x : $\log_3((x+2)(x+4)) + \log_{1/3}(x+2) < \frac{1}{2}\log_{\sqrt{3}} 7$.

Solution : Checking existance,

$(x+2)(x+4) > 0$ and $(x+2) > 0$

$x < -4$ or $x > -2$ and $x > -2$

Now solving inequality.

$$\log_3((x+2)(x+4)) + \log_{1/3}(x+2) < \frac{1}{2}\log_{\sqrt{3}} 7.$$

$\Rightarrow \log_3(x+2)(x+4) - \log_3(x+2) < \log_3 7$

$\Rightarrow \log_3(x+4) < \log_3 7$

$\Rightarrow x + 4 < 7 \Rightarrow x < 3$

$\Rightarrow -2 < x < 3 \Rightarrow \quad x \in (-2, 3)$

Illustration 24 : Solve for x : $\log_{1/3}\log_4(x^2-5)>0$

Solution : Checking existence,

(i) $\log_4(x^2-5)>0 \Rightarrow x^2-5>1$

$$\Rightarrow \left(x-\sqrt{6}\right)\left(x+\sqrt{6}\right)>0$$

$$\Rightarrow x \in \left(-\infty,-\sqrt{6}\right)\cup\left(\sqrt{6},\infty\right)$$

(ii) $x^2-5>0 \Rightarrow x \in \left(-\infty,-\sqrt{5}\right)\cup\left(\sqrt{5},\infty\right)$

solving inequality,

$$\log_{\frac{1}{3}}\log_4\left(x^2-5\right)>0$$

$$\Rightarrow \log_4\left(x^2-5\right)<1$$

$$\Rightarrow x^2-5<4 \Rightarrow x^2-9<0$$

$$\Rightarrow (x-3)(x+3)<0$$

$$\Rightarrow x \in (-3,3) \qquad \text{...(iii)}$$

$\therefore$ Answer : (i) $\cap$ (ii) $\cap$ (iii)

$$\Rightarrow x \in \left(-3,-\sqrt{6}\right)\cup\left(\sqrt{6},3\right)$$

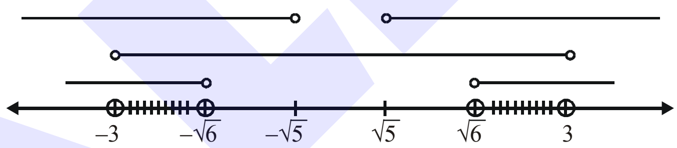


Illustration 25 : Solve for x : $\dfrac{\log_2\left(4x^2-x-1\right)}{\log_2\left(x^2+1\right)}>1$

Solution : Checking existance

(i) $x^2+1\neq 1 \Rightarrow x\neq 0$

(ii) $4x^2-x-1>0$

Now solving inequality,

$$\frac{\log_2\left(4x^2-x-1\right)}{\log_2\left(x^2+1\right)}>1$$

$\Rightarrow \log_2(4x^2 - x - 1) > \log_2(x^2 + 1) \Rightarrow \log_2(4x^2 - x - 1) - \log_2(x^2 + 1) > 0$

$\Rightarrow \log_2 \dfrac{4x^2 - x - 1}{x^2 + 1} > 0$

$\Rightarrow 4x^2 - x - 1 - x^2 - 1 > 0 \Rightarrow 3x^2 - x - 2 > 0 \Rightarrow 3x^2 - 3x + 2x - 2 > 0$

$\Rightarrow 3x(x - 1) + 2(x - 1) > 0 \Rightarrow (x - 1)(3x + 2) > 0 \Rightarrow x < -2/3$ or $x > 1$

$\Rightarrow x \in (-\infty, -2/3)$ or $x \in (1, \infty)$

Note : $x < -\dfrac{2}{3}$ and $x > 1$; $4x^2 - x - 1 > 0$

Illustration 26 : Solve for x : $\log_4\left(3^x - 1\right)\log_{1/4}\left(\dfrac{3^x - 1}{16}\right) \le \dfrac{3}{4}$

Solution : Checking existance $3^x - 1 > 0 \Rightarrow x > 0$,

Now solving inequality

$$\log_4\left(3^x - 1\right)\log_{1/4}\left(\frac{3^x - 1}{16}\right) \le \frac{3}{4}$$

$$\Rightarrow \log_4\left(3^x - 1\right).\left[-\log_4\left(3^x - 1\right) + \log_4 16\right] \le \frac{3}{4}$$

$$\Rightarrow \log_4\left(3^x - 1\right)\left[-\log_4\left(3^x - 1\right) + 2\right] \le \frac{3}{4} \Rightarrow -\left[\log_4\left(3^x - 1\right)\right]^2 + 2\left[\log_4\left(3^x - 1\right)\right] \le \frac{3}{4}$$

Put $\log_4\left(3^x - 1\right) = t \Rightarrow -t^2 + 2t \le \dfrac{3}{4}$

$\Rightarrow -4t^2 + 8t - 3 \le 0 \Rightarrow 4t^2 - 8t + 3 \ge 0 \Rightarrow 4t^2 - 6t - 2t + 3 \ge 0$

$\Rightarrow 2t(2t - 3) - 1(2t - 3) \ge 0 \Rightarrow (2t - 3)(2t - 1) \ge 0$

$\Rightarrow \log_4\left(3^x - 1\right) \le \dfrac{1}{2}$ or $\log_4\left(3^x - 1\right) \ge \dfrac{3}{2}$

$\Rightarrow 0 < 3^x - 1 \le 4^{1/2}$ or $3^x - 1 \ge 4^{3/2}$

$\Rightarrow 1 < 3^x \le 3$ or $3^x \ge 9$

$\Rightarrow 0 < x \le 1$ or $x \ge 2$

$\Rightarrow x \in (0, 1] \cup [2, \infty)$

Illustration 27 : Solve for x : $\log_{1/3}(x^2 - 6x + 18) - 2\log_{1/3}(x-4) < 0$

Solution : Checking existance

(1) $x^2 - 6x + 18 > 0 \Rightarrow x \in \mathbb{R}$

(2) $x - 4 > 0 \Rightarrow x > 4$

Now solving inequality

$\log_{1/3}(x^2 - 6x + 18) - \log_{1/3}(x-4)^2 < 0$

$$\Rightarrow \log_{1/3}\frac{\left(x^2-6x+18\right)}{\left(x-4\right)^2} < 0 \text{ and } 2\log_{1/3}(x-4) = \log_{1/3}(x-4)^2$$

only when $x - 4 > 0$, so we get

$$\log_{1/3}\left(\frac{x^2-6x+18}{(x-4)^2}\right) < 0 \text{ and } x - 4 > 0 \Rightarrow x > 4 \qquad ...(1)$$

$\Rightarrow x^2 - 6x + 18 > (x-4)^2 \Rightarrow x^2 - 6x + 18 > x^2 - 8x + 16$

$2x + 2 > 0 \Rightarrow x > -1 \Rightarrow x \in (-1,\infty)$ ...(2)

from equation (1) and (2), we get $x \in (4,\infty)$

Illustration 28 : Solve for x : $\log_e(x^2 - 2x - 2) \le 0$

Solution : The values of x satisfying the inequality $\log_e(x^2 - 2x - 2) \le 0$ must be such that

$$0 < x^2 - 2x - 2 \le 1$$

we have, $x^2 - 2x - 2 > 0 \Rightarrow (x-1)^2 > 3$

$\Rightarrow \quad |x-1|^2 > 3 \Rightarrow |x-1| > \sqrt{3} \quad \Rightarrow \quad x - 1 > \sqrt{3}$ or $x - 1 < -\sqrt{3}$

$\Rightarrow \quad x > 1 + \sqrt{3}$ or $x < 1 - \sqrt{3}$ ...(1)

Again $x^2 - 2x - 2 \le 1 \Rightarrow x^2 - 2x \le 3 \Rightarrow (x-1)^2 \le 4 \Rightarrow |x-1|^2 \le 4$

$\Rightarrow \quad |x-1| \le 2 \Rightarrow -2 \le x - 1 \le 2 \Rightarrow -1 \le x \le 3$ ...(2)

The value of x satisfying both the inequalities equation (1) and (2) are given by;

Hence, $x \in \left[-1, 1-\sqrt{3}\right) \cup \left(1+\sqrt{3}, 3\right]$

Illustration 29 : Solve for x : $\log_x\left(2x-\frac{3}{4}\right)>2$

Solution : For existanc of logarithm

$$2x-\frac{3}{4}>0 \text{ and } x>0 \text{ and } x\neq 1$$

so, $x\in\left(\frac{3}{8},\infty\right)-\{1\}$

To find the value of x satisfying the inequality $\log_x[2x-(3/4)]>2$

Case I. Let $0<x<1$

Then, $\log_x[2x-(3/4)]>2 \Rightarrow [2x-(3/4)]<x^2$

$\Rightarrow x^2-2x+(3/4)>0 \Rightarrow 4x^2-8x+3>0 \quad \Rightarrow (2x-1)(2x-3)>0$

$\Rightarrow \left(x-\frac{1}{2}\right)\left[x-(3/2)\right]>0 \Rightarrow x>3/2$ or $x<1/2$

$\Rightarrow x<1/2$ because we have $0<x<1$.

$\therefore$ But for $\log[2x-(3/4)]$ to be meaningful, we must have

$2x-(3/4)>0 \Rightarrow x>3/8$

Therefore, if $0<x<1$, the values of x satisfying the given inequality are given by :

$3/8<x<1/2$

Case II. Let $x>1$

Then, $\log_x[2x-(3/4)]>2 \Rightarrow [2x-(3/4)]>x^2$

$\Rightarrow x^2-2x+(3/4)<0 \;\Rightarrow 4x^2-8x+3<0 \;\Rightarrow (2x-1)(2x-3)<0$

$\Rightarrow \left(x-\frac{1}{2}\right)[x-(3/2)]<0 \Rightarrow 1/2<x<3/2$

But we have $x>1$

$\therefore$ We must have $1<x<3/2$ and obviously these values of x make $2x-(3/4)>0$

Thereofre, if $x>1$, the values of x satisfying the given inequality are given by, $1<x<3/2$

$$x\in\left(\frac{3}{8},\frac{1}{2}\right)\cup\left(1,\frac{3}{2}\right)$$

Illustration 30 : Solve for x : $\log_{0.5}(x^2 - 5x + 6) \geq -1$

Solution : Checking existance $x^2 - 5x + 6 > 0$,

$\Rightarrow x \in (-\infty,2) \cup (3,\infty)$

Now solving inequality

$\log_{0.5}(x^2 - 5x + 6) \geq -1 \quad \Rightarrow \quad 0 < x^2 - 5x + 6 \leq (0.5)^{-1}$

$\Rightarrow \quad x^2 - 5x + 6 \leq 2$

$$\begin{cases} x^2 - 5x + 6 > 0 \\ x^2 - 5x + 6 \leq 2 \end{cases} \Rightarrow x \in [1,2) \cup (3,4]$$

Hence, solution set of original inequation : $x \in [1,2) \cup (3,4]$

Illustration 31 : Solve for x : $\log_2 x \leq \dfrac{2}{\log_2 x - 1}$.

Solution : Let $\quad \log_2 x = t$

$$t \leq \frac{2}{t-1} \Rightarrow t - \frac{2}{t-1} \leq 0$$

$$\Rightarrow \quad \frac{t^2 - t - 2}{t-1} \leq 0 \Rightarrow \frac{(t-2)(t+1)}{(t-1)} \leq 0$$

$\Rightarrow \quad t \in (-\infty, -1] \cup (1,2]$

or $\quad \log_2 x \in (-\infty,-1] \cup (1,2]$

or $\quad x \in \left(0, \dfrac{1}{2}\right] \cup (2,4]$

Illustration 32 : Find all x such that $\log_{1/2} x > \log_{1/3} x$.

Solution : We have $\quad \log_{1/2} x > \log_{1/3} x$.

$\Rightarrow \quad -\log_2 x > -\log_3 x \quad \Rightarrow \quad \log_2 x < \log_3 x$

$$\Rightarrow \quad \log_2 x < \frac{\log_2 x}{\log_2 3}$$

$\Rightarrow \quad \log_2 3 \log_2 x < \log_2 x \qquad (\text{as } \log_2 3 > 0)$

$\Rightarrow \quad \log_2 x (\log_2 3 - 1) < 0$

Since $\log_2 3 - 1 > 0$, from the latter inequality we obtain $\log_2 x < 0$, hence $x < 1$. But the original inequality is meaningful only when $x > 0$. Therefore all x that satisfy the original inequality lie in the interval $0 < x < 1$.

Answer : $\quad \mathbf{x \in (0, 1)}$

Illustration 33 : Solve the inequation : $\log_{2x+3} x^2 < \log_{2x+3}(2x+3)$

Solution : For existance of logarithm

(i) $x^2 > 0$ (ii) $2x+3>0$ (iii) $2x+3 \neq 1 \Rightarrow x \in \left(-\frac{3}{2}, \infty\right) - \{-1, 0\}$ ...(i)

Now solving inequality

Case I : $0 < 2x+3 < 1 \Rightarrow -\frac{3}{2} < x < -1$

$\because \log_{2x+3} x^2 < \log_{2x+3} 2x+3$

$\Rightarrow x^2 > 2x+3$

$\Rightarrow (x-3)(x+1) > 0$

$\Rightarrow x \in (-\infty, -1) \cup (3, \infty)$; but $-\frac{3}{2} < x < -1$

$\Rightarrow x \in \left(-\frac{3}{2}, -1\right)$ intersection with (i) $\Rightarrow x \in \left(-\frac{3}{2}, -1\right)$

Case II : 2x + 3 > 1 ⇒ x > –1

$\because \log_{2x+3} x^2 < \log_{2x+3} 2x+3$

$\Rightarrow x^2 < 2x+3$

$\Rightarrow (x-3)(x+1) < 0$

$\Rightarrow x \in (-1, 3)$; but $x > -1$

$\Rightarrow x \in (-1, 3)$ intersection with (i) $\Rightarrow x \in (-1, 3) - \{0\}$

$\therefore$ x $\in$ case I $\cup$ case II

$\Rightarrow x \in \left(-\frac{3}{2}, -1\right) \cup (-1, 0) \cup (0, 3)$

Do yourself-5 :

Solve the following inequalities.

1. $\log_{0.3}(x^2+8) > \log_{0.3}(9x)$ **2.** $\log_7\left(\dfrac{2x-6}{2x-1}\right) > 0$ **3.** $\log_{1/3}\sqrt{x^2-2x} > -\dfrac{1}{2}$

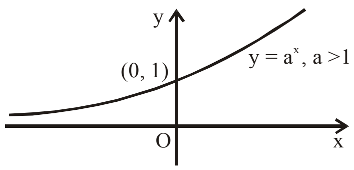
4. $(\log 100x)^2 + (\log 10x)^2 + \log x \le 14$ (Here Base of log is 10)

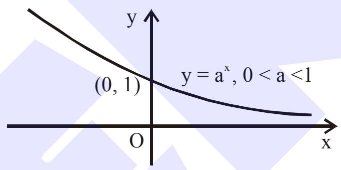
5. $\log_{1/2}(x+1) > \log_2(2-x)$ **6.** $\log_x 2 \,.\, \log_{2x} 2 \,.\, \log_2 4x > 1$

7. $\log_{1/5}(2x^2+5x+1) < 0$ **8.** $\log_x \dfrac{4x+5}{6-5x} < -1$

9. $\log_{x^2}(2+x) < 1$ **10.** $\log_x \dfrac{4x+5}{6-5x} < -1$

7. GRAPH OF EXPONENTIAL FUNCTION, ITS EQUATION AND INEQUALITIES





$$\text{If } a^{f(x)} > b \Rightarrow \begin{cases} f(x) > \log_a b \text{ when } a > 1 \\ f(x) < \log_a b \text{ when } 0 < a < 1 \end{cases}$$

Note : Domain of a^x is $\mathbb{R}$ and Range is $(0, \infty)$.

Illustration 34 : **Solve the equation** $3^x.8^{\frac{x}{x+2}} = 6$.

Solution : Some students solved this equation thus : rewriting it as

$$3^x.2^{\frac{3x}{x+2}} = 3^1.2^1$$

They chose a root x so that the exponents of the respective bases were the same :

$$x = 1, \ \frac{3x}{x+2} = 1$$

hence the "answer" $x = 1$.

But this "answer" is incorrect in the sense that only one root of the equation is found and nothing has been said about any other roots. Actually, if the exponents on the appropriate bases are equal, then the products of these powers are equal, however the converse is not in any way implied and is simply incorrect. For instance, the equation

$$3^1.\ 2^1 = 3^2.\ 2^{\log_2(2/3)}$$

is valid, but $1 \ne 2$ and $1 \ne \log_2(2/3)$. Therefore, the foregoing reasoning may lead to a loss of roots, and this is exactly what occurred in the equation at hand.

Taking logarithms of both members of the original to the base 10, we get

$$x\log_{10}3 + \frac{3x}{x+2}\log_{10}2 = \log_{10}6$$

or $\quad x^2\log_{10}3 + x\,(3\log_{10}2 + 2\log_{10}3 - \log_{10}6) - 2\log_{10}6 = 0$

We now have to solve this quadratic equation. This can be done using a familiar formula, but we will try to simplify the solution by an ingenious device, since we have already seen, by trial and error, that $x_1 = 1$ is a root of the original equation and, consequently, satisfies the equivalent quadratic equation. For this reason, by Viete's theorem the second root of the quadratic equation is $x_2 = (-2\log_{10}6)/\log_{10}3 = -2\log_3 6$ and so the original equation has two roots;

$$x_1 = 1,\ x_2 = -2\log_3 6.$$

Thus, it is useful to be able to guess a root, but never consider the guessing as the whole solution.

Illustration 35 : $\begin{cases} 2^{y-x}(x+y) = 1, \\ (x+y)^{x-y} = 2. \end{cases}$

Solution : The domain of definition is $x + y > 0$.

$$\begin{cases} x + y = \dfrac{1}{2^{y-x}} = 2^{x-y}, \\ (x+y)^{x-y} = 2. \end{cases}$$

From the first equation we find $x + y = 2^{x-y}$ and substitute it into the second equation.

Then $(2^{x-y})^{x-y} = 2 \Rightarrow 2^{(x-y)^2} = 2 \Rightarrow (x-y)^2 = 1$.

The solution of the given system is the solution of the collection of systems

$$\begin{cases} x - y = 1, \\ x + y = 2, \end{cases} \quad \text{or} \quad \begin{cases} x - y = -1, \\ x + y = \dfrac{1}{2}. \end{cases}$$

Answer : $\left(\frac{3}{2}, \frac{1}{2}\right), \left(-\frac{1}{4}, \frac{3}{4}\right)$.

Illustration 36 : Solve for x : $2^{x+2} > \left(\frac{1}{4}\right)^{\frac{1}{x}}$

Solution : We have $2^{x+2} > 2^{-2/x}$. Since the base 2 > 1, we have $x + 2 > -\frac{2}{x}$

(the sign of the inequality is retained).

Now $x+2+\frac{2}{x}>0 \quad \Rightarrow \quad \frac{x^2+2x+2}{x}>0$

$\Rightarrow \quad \frac{(x+1)^2+1}{x}>0 \quad \Rightarrow \quad x \in (0, \infty)$

Illustration 37 : Solve for x : $(1.25)^{1-x} < (0.64)^{2(1+\sqrt{x})}$

Solution : We have $\left(\frac{5}{4}\right)^{1-x} < \left(\frac{16}{25}\right)^{2(1+\sqrt{x})}$ or $\left(\frac{4}{5}\right)^{x-1} < \left(\frac{4}{5}\right)^{4(1+\sqrt{x})}$

Since the base $0<\frac{4}{5}<1$, the inequality is equivalent to the inequality $x-1>4(1+\sqrt{x})$

$\Rightarrow \quad \frac{x-5}{4}>\sqrt{x}$

Now, R.H.S. is positive

$\Rightarrow \quad \frac{x-5}{4}>0 \quad \Rightarrow \quad x > 5$ ....(i)

we have $\frac{x-5}{4}>\sqrt{x}$

both sides are positive, so squaring both sides

$\Rightarrow \quad \frac{(x-5)^2}{16}>x$ or $\frac{(x-5)^2}{16}-x>0$

or $x^2 - 26x + 25 > 0$ or $(x-25)(x-1) > 0$

$\Rightarrow \quad x \in (-\infty, 1) \cup (25, \infty)$ ....(ii)

intersection (i) & (ii) gives $x \in (25, \infty)$

Do yourself - 6 :

Solve for $x \in \mathbb{R}$

1. $4^x - 10.2^{x-1} = 24$

2. $4.2^{2x} - 6^x = 18.3^{2x}$

3. $3^{2x-3} - 9^{x-1} + 27^{2x/3} = 675$

4. $7^{x+2} - \frac{1}{7}.7^{x+1} - 14.7^{x-1} + 2.7^x = 48$

5. $\left(\frac{5}{3}\right)^{x+1} . \left(\frac{9}{25}\right)^{x^2+2x-11} = \left(\frac{5}{3}\right)^9$

6. $\left(3^{x^2-7.2x+3.9} - 9\sqrt{3}\right)\log(7-x) = 0$

7. $5^{2x} = 3^{2x} + 2(5^x) + 2(3^x)$

8. $\left(\frac{2}{3}\right)^{\frac{|x|-1}{|x|+1}} > 1$

9. $(1.25)^{1-(\log_2 x)^2} < (0.64)^{2+\log_{\sqrt{2}} x}$

10. $\left(\frac{1}{2}\right)^{\log_3 \log_{1/5}\left(x^2 - \frac{4}{5}\right)} < 1$

11. $4^x < 2^{x+1} + 3$

12. $e^{\frac{x^2+2}{x^2-1}} < \frac{1}{e^2}$

8. CHARACTERISTIC AND MANTISSA :

For any given number N, logarithm can be expressed as $\log_a N$ = Integer + Fraction

The integer part is called characteristic and the fractional part (always taken non negative) is called mantissa. When the value of $\log_{10} N$ is given, then to find digits of 'N' we use only the mantissa part. The characteristic is used only in determining the number of digits in the integral part (if $N \geq 1$) or the number of zeros after decimal & before first non-zero digit in the number (if $0 < N < 1$).

Note :

(i) The mantissa part of logarithm of a number is always non-negative $(0 \leq m < 1)$

(ii) If the characteristic of $\log_{10} N$ be 'C' and $C \geq 0$, then the number of digits in N is (C + 1)

(iii) If the characteristic of $\log_{10} N$ be '– C' and $C > 0$, then there exist (C – 1) zeros after decimal in N.

In summary, if characteristic of $\log_{10} N$ is 'C' then number of digits ($N \geq 1$, $N \in \mathbb{N}$) or number of zeros after decimal in N ($0 < N < 1$) = $|C + 1|$.

Do yourself - 7 :

1. Evaluate : $\log_{10}(0.06)^6$

2. Find number of digits in 18^{20}

3. Determine number of cyphers (zeros) between decimal & first significant digit in $\left(\frac{1}{6}\right)^{200}$

4. Let **'L'** denotes the antilog of 0.4 to the base 1024.

and **'M'** denotes the number of digits in 6^{10} (Given $\log_{10}2 = 0.3010$, $\log_{10}3 = 0.4771$)

and **'N'** denotes the number of positive integers which have the characteristic 2, when base of the logarithm is 6.

Find the value of LMN.

5. If characteristic of $\log_{10}(0.00006)$ is A & characteristic $\log_3 750$ is B, then A + B is

6. If $\log_{10}2 = .3010$, $\log_{10}3 = .4771$, then number of digits in $4^8.3^7.5^3$ is 'P', then $\frac{P-1}{2}$ is

7. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, then find sum of the number of digits in 6^{15} and the number of zeros immediately after the decimal in 3^{-100}.

EXERCISE (O-1)

Straight Objective Type

1. If $a^4 \cdot b^5 = 1$ then the value of $\log_a(a^5b^4)$ equals

(A) 9/5 (B) 4 (C) 5 (D) 8/5 **LG0142**

2. The expression $\log_p \underbrace{\sqrt[p]{\sqrt[p]{\sqrt[p]{\ldots\ldots\sqrt[p]{p}}}}}_{\text{n radical sign}}$, where $p \ge 2, p \in N ; n \in N$ when simplified is

(A) independent of p (B) independent of p and of n

(C) dependent on both p and n (D) positive **LG0143**

3. The number $\log_2 7$ is

(A) an integer (B) a rational number

(C) an irrational number (D) a prime number

LG0015

4. If a, b, c are positive real numbers such that $a^{\log_3 7} = 27;\ b^{\log_7 11} = 49$ and $c^{\log_{11} 25} = \sqrt{11}$. The value of $\left(a^{(\log_3 7)^2} + b^{(\log_7 11)^2} + c^{(\log_{11} 25)^2}\right)$ equals

LG0074

(A) 489 (B) 469 (C) 464 (D) 400

5. If $2^a = 3$ and $9^b = 4$ then value of (ab) is -

(A) 1 (B) 2 (C) 3 (D) 4

LG0001

6. $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \log_{10}(\log_4 5) + \ldots + \log_{10}(\log_{1023} 1024)$ simplifies to

(A) a composite (B) a prime number

(C) rational which is not an integer (D) an integer **LG0144**

7. If $\log_2(x^2 + 1) + \log_{13}(x^2 + 1) = \log_2(x^2 + 1)\log_{13}(x^2 + 1)$, $(x \ne 0)$, then $\log_7(x^2 + 24)$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

LG0006

8. Given $\log_3 a = p = \log_b c$ and $\log_b 9 = \dfrac{2}{p^2}$. If $\log_9\left(\dfrac{a^4b^3}{c}\right) = \alpha p^3 + \beta p^2 + \gamma p + \delta$ $(\forall\ p \in R - \{0\})$, then $(\alpha + \beta + \gamma + \delta)$ equals

(A) 1 (B) 2 (C) 3 (D) 4

LG0007

9. The greatest value of $(4\log_{10}x - \log_x(.0001))$ for $0 < x < 1$ is -

(A) 4 (B) –4 (C) 8 (D) –8

LG0014

10. If $60^a = 3$ and $60^b = 5$ then the value of $12^{\frac{1-a-b}{2(1-b)}}$ equals

(A) 2 (B) 3 (C) $\sqrt{3}$ (D) $\sqrt{12}$

LG0017

11. Let ABC be a triangle right angled at C. The value of $\frac{\log_{b+c} a + \log_{c-b} a}{\log_{b+c} a \cdot \log_{c-b} a}$ $(b+c \neq 1, c-b \neq 1)$ equals

(A) 1 (B) 2 (C) 3 (D) 1/2

LG0018

12. If α and β are the roots of the equation $(\log_2 x)^2 + 4(\log_2 x) - 1 = 0$ then the value of $\log_\beta \alpha + \log_\alpha \beta$ equals

(A) 18 (B) –16 (C) 14 (D) –18

LG0019

13. If $\log_2\left(4+\log_3(x)\right) = 3$, then sum of digits of x is -

(A) 3 (B) 6 (C) 9 (D) 18

LG0002

14. Sum of all the solution(s) of the equation $\log_{10}(x) + \log_{10}(x+2) - \log_{10}(5x+4) = 0$ is -

(A) –1 (B) 3 (C) 4 (D) 5

LG0003

15. The product of all the solutions of the equation $x^{1+\log_{10} x} = 100000x$ is -

(A) 10 (B) 10^5 (C) 10^{-5} (D) 1

LG0004

16. If x_1 and x_2 are the roots of equation $e^{3/2}.\ x^{2\ell nx} = x^4$, then the product of the roots of the equation is -

(A) e^2 (B) e (C) $e^{3/2}$ (D) e^{-2}

LG0005

17. If $\log_a\left(1-\sqrt{1+x}\right) = \log_{a^2}\left(3-\sqrt{1+x}\right)$, then number of solutions of the equation is -

(A) 0 (B) 1 (C) 2 (D) infinitely many

LG0008

18. The number of solution(s) of $\sqrt{\log_3(3x^2).\log_9(81x)} = \log_9 x^3$ is -

(A) 0 (B) 1 (C) 2 (D) 3

LG0009

19. If $x, y \in 2^n$ when $n \in I$ and $1 + \log_x y = \log_2 y$, then the value of $(x + y)$ is -

(A) 2 (B) 4 (C) 6 (D) 8

LG0011

20. The number of integral solutions of $|\log_5 x^2 - 4| = 2 + |\log_5 x - 3|$ is -

(A) 1 (B) 2 (C) 3 (D) 0

LG0016

21. If $\log_{10}\left(\frac{1}{2^x+x-1}\right)=x\left(\log_{10}5-1\right)$, then $x=$

(A) 4 (B) 3 (C) 2 (D) 1

LG0021

22. The number of solutions of the equation $\log_{x-3}(x^3-3x^2-4x+8)=3$ is equal to

(A) 4 (B) 3 (C) 2 (D) 1

LG0022

23. Sum of the roots of the equation $9^{\log_3(\log_2 x)}=\log_2 x-(\log_2 x)^2+1$ is equal to

(A) 2 (B) 4 (C) 6 (D) 8

LG0023

24. If 1, $\log_9(3^{1-x}+2)$ and $\log_3(4.3^x-1)$ are in A.P. then x can be

(A) $\log_4 3$ (B) $\log_3 4$ (C) $1-\log_3 4$ (D) $\log_3 0.25$

LG0025

25. Number of real solutions of the equation $\sqrt{\log_{10}(-x)}=\log_{10}\sqrt{x^2}$ is :

(A) zero (B) exactly 1 (C) exactly 2 (D) 4 **LG0145**

26. If x_1 & x_2 are the two values of x satisfying the equation $7^{2x^2}-2\left(7^{x^2+x+12}\right)+7^{2x+24}=0$, then (x_1+x_2) equals -

(A) 0 (B) 1 (C) –1 (D) 7

LG0010

27. If $\log_{0.3}(x-1)<\log_{0.09}(x-1)$, then x lies in the interval

(A) $(2,\infty)$ (B) $(1,2)$ (C) $(1,\infty)$ (D) none of these

LG0020

28. If x satisfies the inequality $\log_{25}x^2+(\log_5 x)^2<2$, then $x\in$

(A) $\left(\frac{1}{25},5\right)$ (B) $(1,2)$ (C) $(4,5)$ (D) $(0,1)$

LG0024

29. If $x=\log_2\left(\sqrt{56+\sqrt{56+\sqrt{56+\sqrt{56+.......\infty}}}}\right)$, then which of the following statements holds good ?

(A) $x<0$ (B) $0<x<2$ (C) $2<x<4$ (D) $3<x<4$

LG0013

30. If $n\in N$ such that characteristic of n^2 to the base 8 is 2, then number of possible values of n is -

(A) 14 (B) 15 (C) 448 (D) infinite

LG0012

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $\log_a x = b$ for permissible values of a and x then identify the statement(s) which can be correct ?

(A) If a and b are two irrational numbers then x can be rational.

(B) If a rational and b irrational then x can be rational.

(C) If a irrational and b rational then x can be rational.

(D) If a rational and b rational then x can be rational. **LG0146**

2. Which of the following statements is(are) correct ?

(A) $7^{1/7} > (42)^{1/14} > 1$ (B) $\log_3(5)\, \log_7(9)\, \log_{11}(13) > -2$

(C) $\sqrt{99+70\sqrt{2}} + \sqrt{99-70\sqrt{2}}$ is rational (D) $\dfrac{1}{\log_4 3} + \dfrac{1}{\log_7 3} > 3$

LG0027

3. Let $N = \dfrac{\log_2 24}{\log_{96} 2} - \dfrac{\log_2 192}{\log_{12} 2}$. Then N is : **LG0147**

(A) a natural number (B) a prime number (C) a rational number (D) an integer

4. If $\left(\log_\beta \alpha\right)^2 + \left(\log_\alpha \beta\right)^2 = 79$, $(\alpha > 0, \beta > 0, \alpha \neq 1, \beta \neq 1)$ then value of $(\log_\beta \alpha) + (\log_\alpha \beta)$ can be -

(A) 7 (B) –9 (C) 9 (D) –7

LG0026

5. For $a > 0, \neq 1$ the roots of the equation $\log_{ax} a + \log_x a^2 + \log_{a^2x} a^3 = 0$ can be

(A) $a^{-3/4}$ (B) $a^{-4/3}$ (C) $a^{-1/2}$ (D) none of these

LG0028

6. The solution of the equation $5^{\log_a x} + 5x^{\log_a 5} = 3$, $(a > 0, a \neq 1)$ is

(A) $a^{-\log_5 2}$ (B) $a^{\log_5 2}$ (C) $2^{-\log_5 a}$ (D) $2^{\log_5 a}$

LG0029

7. The equation $x^{(3/4)(\log_2 x)^2 + \log_2 x - 5/4} = \sqrt{2}$ has

(A) at least one real solution (B) exactly three real solutions

(C) exactly one irrational solution (D) real + non real roots

LG0031

8. The equation $\log_{x^2} 16 + \log_{2x} 64 = 3$ has :

(A) one irrational solution (B) no prime solution

(C) two real solutions (D) one integral solution **LG0148**

9. The equation $x^{\left[(\log_3 x)^2 - \frac{9}{2}\log_3 x + 5\right]} = 3\sqrt{3}$ has

(A) exactly three real solution (B) at least one real solution

(C) exactly one irrational solution (D) complex roots **LG0149**

10. Let a and b be real numbers greater than 1 for which there exists a positive real number c, different from 1, such that $2(\log_a c + \log_b c) = 9\log_{ab} c$, then the possible value of $\log_a b$.

(A) 1/2 (B) 3 (C) 1/3 (D) 2

LG0071

11. If x_1, x_2 are the value(s) of x satisfying the equation $\log_2^2(x-2) + \log_2(x-2)\log_2\left(\frac{3}{x}\right) - 2\log_2^2\left(\frac{3}{x}\right) = 0$, then $x_1 + x_2$ is equal to

(A) $x_1 + x_2 = 9$ (B) $x_1 x_2 = 18$ (C) $|x_1 - x_2| = 3$ (D) $x_1^2 + x_2^2 = 45$

LG0040

12. The solution set of the system of equations $\log_3 x + \log_3 y = 2 + \log_3 2$ and $\log_{27}(x+y) = \frac{2}{3}$ is :

(A) {6, 3} (B) {3, 6} (C) {6, 12} (D) {12, 6} **LG0151**

13. If $2^{x+y} = 6^y$ and $3^{x-1} = 2^{y+1}$, then the value of $(\log 3 - \log 2)/(x - y)$ is

(A) 1 (B) $\log_2 3 - \log_3 2$ (C) log(3/2) (D) log 3 – log 2

LG0030

14. The inequality $-1 \le \left(\frac{1}{3}\right)^x < 2$ is satisfied by

(A) $x \in [0, 1]$ (B) $x > -\log_3 2$ (C) $x < -\log_3 2$ (D) $x < -1$

LG0032

15. Which of the following statements are true

(A) $\log_2 3 < \log_{12} 10$ (B) $\log_6 5 < \log_7 8$

(C) $\log_3 26 < \log_2 9$ (D) $\log_{16} 15 > \log_{10} 11 > \log_7 6$ **LG0150**

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 and 2

Let $\log_2 5 = \mathbf{a}$; $\log_5 3 = \mathbf{b}$

1. The value of $\log_{50} 30$ is equal to

(A) $\frac{b+ab+1}{b+2}$ (B) $\frac{a+ab+1}{2a+1}$ (C) $\frac{a+ab+1}{a+2}$ (D) $\frac{a+ab+b}{2a+1}$

LG0033

2. Roots of the equation $\log_{10}2.\log_{10}5x^2 - (\log_{10}5 + \log_{10}15.\log_{10}2)x + \log_{10}15 = 0$ are

(A) **a, b** (B) $1 + \mathbf{a}$, $1 + \mathbf{b}$ (C) $\mathbf{a} - 1$, $1 - \mathbf{b}$ (D) $\mathbf{a} + 2$; $\mathbf{b} + 2$

LG0033

Paragraph for Question 3 and 4

Let $\log_5 N = I_1 + f_1$ and $\log_3 N = I_2 + f_2$, where I_1, I_2 are integers and f_1, $f_2 \in [0,1)$.

On the basis of above information, answer the following questions :

3. If $I_1 = 2$ and $I_2 = 3$, then maximum integral value of N is greater than or equal to -

(A) 79 (B) 80 (C) 81 (D) 82 **LG0152**

4. If $I_1 = 3$ and $I_2 = 4$, then number of possible integral values of N is less than or equal to -

(A) 119 (B) 116 (C) 117 (D) 118 **LG0153**

Paragraph for Question 5 and 6

If (x_1,y_1) and (x_2,y_2) are the solution of the system of equation.

$$\log_{225}(x) + \log_{64}(y) = 4$$

$$\log_x(225) - \log_y(64) = 1,$$

5. The value of $\log_{30}(x_1y_1x_2y_2)$ is **LG0154**

6. Number of digits in $(x_1y_1x_2y_2)$ is

LG0097

Matrix Match Type

7. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	The value of x for which $2\ell n\left(2^x-5\right)=\ell n2+\ell n\left(2^x-\frac{7}{2}\right)$ is	(1)	2
		(2)	4
(Q)	The sum of the real roots of the equation $x^{\log_{10}\left(\frac{5x}{2}\right)}=10^{\log_{10}x}$ is	(3)	3
		(4)	5
(R)	If $\alpha=2^{\frac{(\log_3(\log_3 5))}{\log_3 2}}$ and $\beta^\alpha=9$, then value $\frac{1}{5}\left(\beta^{\alpha^2}\right)$ is equal to	(5)	$\frac{1}{2}$
(S)	$\log_{\sqrt{3}}\left(\sqrt{6+\sqrt{6+\sqrt{6+\ldots\infty}}}\right)$ is equal to		

The correct option is

(A) P → 1; Q → 2; R → 2; S → 5
(B) P → 3; Q → 4; R → 2; S → 5
(C) P → 1; Q → 2; R → 4; S → 1
(D) P → 3; Q → 4; R → 4; S → 1

LG0155

8. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	Positive value of x satisfying the equation $(25)^{\frac{2x^2-5x-9}{4}}=\left(\sqrt{11}\right)^{3\log_{11}5}$ is	(1)	5
(Q)	Sum of values of x which satisfy the equation $\sqrt{\lvert x-4\rvert^{x-2}}=\sqrt[4]{\lvert x-4\rvert^{x+2}}$ is	(2)	12
(R)	The expression $\sqrt{\log_{0.5}^2 32}$ has the value equal to	(3)	4
(S)	The value of 3k, where $\log(\log 9)+\log(\log 49)=\log k+\log(\log 3)+\log(\log 7)$, is	(4)	18

(A) P → 3; Q → 4; R → 1; S → 2
(B) P → 1; Q → 4; R → 3; S → 2
(C) P → 2; Q → 4; R → 3; S → 1
(D) P → 3; Q → 1; R → 2; S → 4

LG0156

9. In the following matrix match **Column-I** has some quantities and **Column-II** has some comments or other quantities

Match the each element in **Column-I** with corresponding element(s) in **Column-II**

Column-I	Column-II
(A) $\log_{\sin\pi/6} e$	(P) $x^{\log_x 2}$; $x > 0$; $x \neq 1$
(B) 2	(Q) $2^{\log_5 7}$
(C) $\log_{\tan\pi/3} \pi$	(R) negative
(D) $7^{\log_5 2}$	(S) positive

LG0157

10.

Column-I	Column-II
(A) Anti logarithm of $(0.\overline{6})$ to the base 27 has the value equal to	(p) 5
(B) Characteristic of the logarithm of 2008 to the base 2 is	
(C) The value of b satisfying the equation, $\log_e 2 \cdot \log_b 625 = \log_{10} 16 \cdot \log_e 10$ is	(q) 7
(D) Number of naughts after decimal before a significant figure comes in the number $\left(\frac{5}{6}\right)^{100}$, is	(r) 9
	(s) 10

(Given $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$)

EXERCISE (O-4)

Numerical Grid Type

1. Let A denotes the value of $\log_{10}\left(\frac{ab+\sqrt{(ab)^2-4(a+b)}}{2}\right)+\log_{10}\left(\frac{ab-\sqrt{(ab)^2-4(a+b)}}{2}\right)$

when a = 43 and b = 57 and B denotes the value of the expression $\left(2^{\log_6 18}\right).\left(3^{\log_6 3}\right)$.

Find the value of (A.B).

LG0045

2. Let $y=\sqrt{\log_2 3.\log_2 12.\log_2 48.\log_2 192+16}-\log_2 12.\log_2 48+10$. Find $y \in N$.

LG0089

3. There exist positive integers A, B and C with no common factor greater than 1 such that $A\log_{200}5 + B\log_{200}2 = C$, then the value of A + B + C is

LG0043

4. If $a = \log_{12}18$ & $b = \log_{24}54$ then find the value of $ab + 5(a - b)$.

LG0077

5. Number of solutions of $\log_{(1+x)}\left(\frac{1-x}{1+x}\right)-2\log_{(1-x)}(1+x)=0$ is

LG0038

6. The minimum possible real x which satisfy the equation, $2\log_2\log_2 x+\log_{1/2}\log_2\left(2\sqrt{2}x\right)=1$.

LG0044

7. Find the square of the sum of the solution of the equation

$\log_3x.\log_4x.\log_5x = \log_3x.\log_4x + \log_4x.\log_5x + \log_5x.\log_3x$.

LG0048

8. If 'x' and 'y' are real numbers such that $2\log(2y - 3x) = \log x + \log y$, find $\frac{y+x}{y-x}$.

LG0075

9. If $4^A + 9^B = 10^C$, where $A = \log_{16}4$, $B = \log_3 9$ & $C = \log_x 83$, then find x. **LG0158**

10. Find the number of positive integers not satisfying the inequality $\log_2(4^x - 2.2^x + 17) > 5$. **LG0159**

EXERCISE (JA)

1. Number of solutions of $\log_4(x-1) = \log_2(x-3)$ is **[JEE 2001 (Screening)]**

(A) 3 (B) 1 (C) 2 (D) 0

LG0106

2. Let (x_0, y_0) be the solution of the following equations **[JEE 2011, 3 (–1)]**

$$(2x)^{\ln 2} = (3y)^{\ln 3}$$

$$3^{\ln x} = 2^{\ln y}$$

Then x_0 is

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) 6

LG0107

3. The value of $6+\log_{\frac{3}{2}}\left(\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\sqrt{4-\frac{1}{3\sqrt{2}}\ldots\ldots}}}\right)$ is **[JEE 2012, 4M]**

LG0108

4. If $3^x = 4^{x-1}$, then $x =$ **[JEE-Advanced 2013, 4, (–1)]**

(A) $\frac{2\log_3 2}{2\log_3 2-1}$ (B) $\frac{2}{2-\log_2 3}$ (C) $\frac{1}{1-\log_4 3}$ (D) $\frac{2\log_2 3}{2\log_2 3-1}$

LG0109

5. The value of $\left((\log_2 9)^2\right)^{\frac{1}{\log_2(\log_2 9)}} \times \left(\sqrt{7}\right)^{\frac{1}{\log_4 7}}$ is ______ **[JEE(Advanced)-2018]**

LG0110

6. The product of all positive real values of x satisfying the equation $x^{\left(16(\log_5 x)^3-68\log_5 x\right)} = 5^{-16}$ is______.

[JEE(Advanced)-2022]

LG0160

ANSWER KEY

Do yourself - 1

1. **(a)** $\log_3 81 = 4$ **(b)** $\log_{10}(0.001) = -3$ **(c)** $\log_{128} 2 = 1/7$

2. **(a)** $32 = 2^5$ **(b)** $4 = (\sqrt{2})^4$ **(c)** $0.01 = 10^{-2}$

3. 6

4. (i) 0 (ii) 1 (iii) 2 (iv) 3 (v) –1
(vi) –5 (vii) –4 (viii) 1/2 (ix) 1 (x) 3/2
(xi) –1/5 (xii) –3/7

5. (i) 0 (ii) 1 (iii) 3 (iv) –4 (v) –1/2
(vi) 1/2 (vii) –3/2 (viii) 9/4

6. (i) 0 (ii) 1 (iii) 2 (iv) 4 (v) –1
(vi) 1/2 (vii) –3/2 (viii) 7/2 (ix) 2/7

7. (i) 0 (ii) 1 (iii) 2 (iv) –1 (v) –2
(vi) –4 (vii) –1/3 (viii) 1/7 (ix) –5/2 (x) 9/4

8. (i) 0 (ii) 1 (iii) 2 (iv) 4 (v) –1
(vi) –2 (vii) –1/2 (viii) 1/4 (ix) 1/2 (x) 1/3
(xi) 1/3

9. (i) 4 (ii) 2 (iii) $a > 0, a \neq 1$ (iv) –5,2 (v) –2,2
(vi) –3,3 (vii) 9 (viii) 1/81 (ix) 1 (x) 2
(xi) $\sqrt{2}, -\sqrt{2}$

10. (i) $\frac{-1}{2}$ (ii) 2 (iii) 1 (iv) 1 **(v)** –1

Do yourself - 2

1. **(a)** 1 **(b)** $\frac{1}{5}$ **2.** 3

4. (i) 2 (ii) 3 (iii) 1 (iv) 4

Do yourself - 3

1. **(i)** 3 **(ii)** 5/6 **(iii)** 0 **(iv)** 2 **(v)** (A)
(vi) (C)

2. (i) 3 (ii) 9 (iii) 9 (vi) 2

3. $2s + 10s^2 - 3(s^3 + 1)$ **5.** $\log_b N$ **6.** 4 **7.** 9

8. 1 **9.** 6 **10.** $\frac{25}{2}$ **11.** 1/6 **12.** 34

13. 32

Do yourself - 4

1. (i) $\sqrt{2},-\sqrt{2}$ (ii) 1 (iii) 1 (iv) 0 (v) 1, –1

2. (i) $x=\pm 2$ (ii) $x=\pm\frac{1}{2}$ (iii) $x=\frac{3}{2}$ (iv) $x=4$

3. $\log_2 3$ 4. $\frac{1}{8},2$ 5. $9,\frac{1}{9}$ 6. 2, 8 7. 1, 0

8. ϕ 9. $100,\frac{1}{100}$ 10. $2^{-\log_3 a}$ 11. ϕ 12. 2

13. $x=8,\frac{1}{8}$ 14. $\{1+\sqrt{3}\}$ 15. {3} 16. {4} 17. {2}

18. {0} 19. {–5} 20. {3/2, 10} 21. $\{10^{-1}, 10\}$ 22. $\{\sqrt{5}, 5\}$

23. $\{-(3-\sqrt{3})/3, 8\}$

Do yourself-5

1. $x \in (1,8)$ 2. $x \in (-\infty, 1/2)$ 3. $(-1,0)\cup(2,3)$

4. $\frac{1}{\sqrt{10^9}} \le x \le 10$ 5. $-1 < x < \frac{1-\sqrt{5}}{2}$ or $\frac{1+\sqrt{5}}{2} < x < 2$

6. $2^{-\sqrt{2}} < x < 2^{-1}$; $1 < x < 2^{\sqrt{2}}$ 7. $(-\infty, -2.5) \cup (0, \infty)$

8. $\frac{1}{2} < x < 1$ 9. $(-2,-1)\cup(-1,0)\cup(0,1)\cup(2,\infty)$ 10. $\left(\frac{1}{2},1\right)$

Do yourself - 6

1. $x=3$ 2. $x=-2$ 3. $x=3$ 4. $x=0$

5. $x=\frac{-7}{2},2$ 6. $x=\frac{1}{5},6$ 7. $x=1$ 8. $x\in(-1,1)$

9. $\left(0,\frac{1}{2}\right)\cup(32,\infty)$ 10. $\left(-1,-\frac{2}{\sqrt{5}}\right)\cup\left(\frac{2}{\sqrt{5}},1\right)$ 11. $(-\infty, \log_2 3)$ 12. $(-1,0)\cup(0,1)$

Do yourself - 7

1. –7.3314 2. 26 3. 155 4. 23040 5. 1

6. 5 7. 59

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	C	B	A	D	B	C	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	C	C	D	A	A	B	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	D	A	C	C	B	A	A	C	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,B,D	A,B,C,D	B,C	B,C	A,C	A,B,C,D	A,B,C,D	A,B,C,D	A,D
Que.	11	12	13	14	15					
Ans.	A,B,C,D	A,B	C,D	A,B	B,C					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	
Ans.	B	B	A,B	A,D	12.00	18.00	D	A	
Q.9	A	B	C	D	Q.10	A	B	C	D
	R	P,S	S	Q,S		r	s	p	q

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	12	6	6	1	0	8.00	3721.00	2.60	10.00	2.00

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6
Ans.	B	C	4	A,B,C	8	1

Important Notes